



Francesco Pelosin



11/02/1992



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Italian

Profile

Hi! I'm Francesco, I'm a Ph.D. in Artificial Intelligence (specifically Continual Learning) and Computer Vision. My passion is applying my knowledge in the industry world to create new technology pipelines. In a nutshell I'm a diligent worker and a problem solver. I'm very quick at developing proof of concepts. I possess good communication skills, also in international environments. I'm naturally inclined toward curiosity and I like to try new things, BUT I always make sure things works. I'm a sensible person and I do my best to maintain a gentle working environment. Although I'm very independent I also enjoy collaborate in a team. My core expertise lies in Continual Learning, Semantic Segmentation, Generative Modeling and Classification for Computer Vision problems.

Education

2018 - 2022

Ph.D. Computer Vision/Continual Learning,

Italy/Spain

Ca' Foscari University, Italy / Computer Vision Center, Spain

Topics: Machine Learning, Deep Learning, Incremental Learning

Supervisor Ita: Prof. **Andrea Torsello**

PI Spain: **Joost van de Weijer**

I collaborated for 9 months (~3 months in-site + 6 remotely) at the Computer Vision Center (CVC) Barcelona, Spain working on continual learning for computer vision. Here the [link](#) to my thesis

2015 - 2018

M.Sc. AI - Data Management and Analytics, Ca' Foscari University

Italy

Final Score: 110L (Full Marks with Distinction).

The thesis has been published in Pattern Recognition (Q1 Journal).

2011 - 2015

B.Sc. Computer Science, Ca' Foscari University

Italy

Final Score: 99

Thesis: I developed a Computer Vision Optical Modem to perform camera calibration from distance

Work Experience

10/2023 - present

Deep Learning Research Engineer, Covision Lab

Brixen, Italy

Role:

I'm a project lead for of several projects with international customers (Pirelli, Microtec, Durst etc). My role is to quickly understand their problem, get a grasp on the data and develop a custom pipeline. I'm also part of the core R&D team which publishes on top venues.

Main projects:

- Prototyped in 4 months a computer vision system based on self-supervised learning for semantic segmentation. The pipeline allows our customer to decrease labeling costs up to 95%. We crafted also a paper based on the project that has been accepted to ECCV2024 VISION workshop organized by Apple.
- I prototyped a generative architecture for synthetic data creation under cheap human annotations. The POC has been delivered to the customer and abstracted into a paper accepted at CVPR Workshops 2025 in Nashville USA.
- I prototyped in 6 months a generative pipeline with high quality fidelity to real world data for Pirelli tyres. The generative pipeline is used for synthetic data creation and later for anomaly detection. I can't disclose on the resume the details.

Side projects:

Provided deep learning routines to create new auto-labeled datasets for Durst.

Soft skills:

I co-advise the selection of new candidates that will join the team as Deep Learning engineers. For now I interviewed together with the CTO three persons on the team which have later joined.

12/2025

Guest Lecturer, Free University of Bozen/Bolzano

Italy

I've been invited to deliver two lectures at the M. Sc. in Computer Engineering held by the Prof. Oswald Lanz on the topic of Transformers and Masked Autoencoders and Diffusion Models. I've prepared slides and a hands-on laboratory.

10/2022 – 10/2023

AI Research Scientist, Leonardo S.p.a.

Rome, Italy

Role:

I was part of the AI R&D team of Leonardo Labs, the defense company of the Italian government. My duty was developing prototypes (up to TRL 4).

Main Project:

During my experience I focused on building a continual learning strategy based on neural network submodules. The work has been accepted as oral to AAAI2024 in Vancouver (Canada). We also prototyped a streamlit app for Automatic Image Annotation exploiting CLIP embeddings as core mechanism. Finally we developed a new methodology for Machine Unlearnin, this work has been kickstarted by me but soon after I left the company

2018

**Computer Vision Research Engineer,
Digital Strategy Innovation s.r.l.**

Venice, Italy

For 6 months, I worked on an industrial project for a big GDO company (PAM Panorama). The project was meant to create a computer vision system based on YOLO detecting out-of-stock products in supermarket shelves.

2018 – 2022

Teaching Assistant / Tutor, Ca' Foscari University

Italy

For several years I was teaching assistant for the course of Computer Architecture, I tutor students, create and correct exams.

Skills

Deep Learning / ML Tools

Pytorch, Jupyter, sklearn, numpy, wandb, dvc, huggingface (basic)...

Languages

Python (proficient 5+ years), C (very basic), C++ (very basic), AI Prompting Patterns

Others

Hiring, Candidate Interviews, Agentic Coding

AI Frameworks

docker, streamlit, git, conda, virtualenv, vim, tmux

Interested in

LLMs, Word Models, Cybersecurity

DL Techniques / Models

Semantic Segmentation, Synthetic Data Generation, Classification, Transformers (swin, vits), Convolutional Neural Networks, Distillation, Masked Autoencoding, Diffusion, Flow Matching

Awards

21/06/2022

Runner-Up Paper Award, Continual Learning CVPR 2022 Workshop

Our paper Towards Exemplar-Free Continual Learning in Vision Transformers: An Account of Attention, FUnctional and Weight Regularization won the runner-up paper award in CLVISION workshop at CVPR 2022 New Orleans, USA

2020

Winner European Copernicus Hackathon, European Commission

We won the possibility to join the European accelerator for startups and a money prize. Here's a video of a cophack. We were awarded by Giorgio Saccoccia, president of Italian Space Agency (ASI) and relevant figures in the space economy scene. Then, we embraced the journey to join the startup accelerator.

2017

1st Place - Hacking Competition (CTF), Ca' Foscari University

During my M.Sc. I won the internal hacking competition held as a course. <Here> some live asciicasts of me hacking into enemy machines (the writeups are hidden, but I have them). Then, I got recruited to join the cybersecurity research group <c00kies@venice> and participate to international capture the flag (CTF) competitions (FAUST 2017).

2023

Breast Cancer Detection Challenge - Bronze Medal, Kaggle

The goal of this competition was to identify breast cancer through a model trained with screening mammograms obtained from regular screening. The competition was hosted, by the Radiological Society of North America (RSNA) a non-profit organization through Kaggle. The certificate of the competition is at the end of the resume.

Publications (first authors are underlined)

06/2025

Bounding Box-Guided Diffusion for Synthesizing Industrial Images and Segmentation Maps, Poster - Synthetic Data Workshop (CVPR 2025)

Alessandro Simoni, Francesco Pelosin

Dynamic Label Injection for Imbalanced Industrial Defect Segmentation, Poster - VISION Workshop (ECCV 2024)

Emanuele Caruso, Francesco Pelosin, Alessandro Simoni, Marco Boschetti

DUCK: Distance-based Unlearning via Centroid Kinematics, arxiv

Marco Cotogni, Jacopo Bonato, Luigi Sabetta, Francesco Pelosin, Alessandro Nicolosi

MIND: Multi-Task Incremental Network Distillation,

Oral - Association for the Advancement of Artificial Intelligence (AAAI 2023)

Jacopo Bonato, Francesco Pelosin, Luigi Sabetta, Alessandro Nicolosi

Simpler Is Better: off-the-shelf Continual Learning through Pretrained Backbones,

Poster - Workshop Transformers for Vision T4F (CVPR 2022)

Francesco Pelosin

Towards Exemplar-Free Continual Learning in Vision Transformers: an Account of Attention, Functional and Weight Regularization,

Poster - Workshop of Continual Learning (CVPR 2022)

Francesco Pelosin, Saurav Jha, Andrea Torsello, Bogdan Raducanu, Joost van de Weijer

Smaller Is Better: an analysis of quality/quality trade-off in rehearsal continual learning, Oral - World Conference of Computational Intelligence 2022 - Proceedings in IJCNN

Francesco Pelosin, Andrea Torsello

Incremental Automatic Image Annotation,

Ital-IA 2023: 3rd National Conference on Artificial Intelligence - Industrial Track

Luigi Sabetta, Francesco Pelosin, Giulia Denevi, Alessandro Nicolosi

Unsupervised semantic discovery through visual patterns detection, S+SSPR, 2021

Francesco Pelosin, Andrea Gasparetto, Andrea Albarelli, Andrea Torsello

Separating Structure from Noise in Large Graphs Using the Regularity Lemma,

Pattern Recognition, 2020

Marco Fiorucci, Francesco Pelosin, Marcello Pelillo

Projects

04/2025 - 05/2025

PodXiv, Podcast generator from any paper

PodXiv is an AI-powered app hosted in Huggingface.

It turns any academic PDF into a conversational podcast between a researcher and a curious student. The name is inspired by arXiv, Alphaxiv, but it is designed to work with any paper or PDF resource. PodXiv helps make complex research accessible and listenable. I made this app to speed up reading papers, I learn faster when I listen to people talking.

You can try it here! <https://huggingface.co/spaces/lakj7/podXiv>

Blog Post, Ephemeral Conda Environments

Short quarto blogpost talking about a quick hack to speed up productivity without polluting the coding environment, while experimenting with models.

You can read it here: <https://lakj.quarto.pub/ephemeral-conda-environments/>